LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Unit Test: Force and Laws of Motion

Chapter / Topic: Force and Laws of Motion (topic-based)

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SECTION A

Q1. The SI unit of force is the: **[1]**

Answer: (c) newton (1 N = 1 kg m/s^2)

Q2. The mass of an object is a measure of its: **[1]**

Answer: (d) inertia

Q3. The working of a rocket is based on: **[1]**

Answer: (a) Newton's third law

Q4. A force of 10 N acts on a mass of 2 kg. The acceleration produced is: **[1]**

Answer: (b) 5 m/s^2

Q5. When a stationary bus starts suddenly, the standing passengers tend to fall: **[1]**

Answer: (c) backward (inertia of rest)

Q6. **Assertion (A):** Action and reaction forces do not cancel each other. **Reason (R):** Action and reaction act on two different bodies. **[1]**

Answer: (a)

SECTION B

Q7. Why is it easier to push an empty box than a box full of books? **[2]**

Answer: The empty box has less mass, and so less inertia. Less force is needed to change its state of rest.

- Less mass: 1
- Less inertia, so less force: 1

OR, answer: The carpet suddenly moves when beaten, but the dust particles tend to stay at rest because of inertia. So they separate from the carpet and fall out.

- Carpet moves: 1
- Dust stays at rest (inertia): 1

Q8. Find the momentum of (a) a 5 kg ball moving at 2 m/s, and (b) a 50 g bullet moving at 200 m/s. Which has more momentum? **[2]**

Answer: (a) p = 5 × 2 = 10 kg m/s. (b) p = 0.05 × 200 = 10 kg m/s. They have equal momentum.

- Each momentum: ½ + ½
- Equal, with 50 g → 0.05 kg: 1

SECTION C

Q9. A car of mass 1000 kg moving at 36 km/h is brought to rest in 5 s. **[3]**

(i) u = 36 km/h = 10 m/s. p = 1000 × 10 = 10,000 kg m/s [1]

(ii) F = change in momentum/time = 10,000/5 = 2000 N (opposite to the motion) [1]

(iii) When the car stops suddenly, passengers keep moving forward because of inertia. Seat belts hold them back and prevent injury. [1]

Q10. State Newton's second law of motion and use it to show that F = ma. **[3]**

Answer: The rate of change of momentum of an object is proportional to the applied force, and takes place in the direction of the force. If a mass m changes velocity from u to v in time t, F ∝ (mv – mu)/t = m(v – u)/t = ma. So F = kma, and the unit of force is chosen so that k = 1, giving F = ma.

- Statement: 1
- Rate of change of momentum: 1
- F = ma: 1

OR, answer: Before firing, the gun and bullet are at rest, so the total momentum is zero. After firing, the bullet moves forward with momentum $m_b v_b$. To keep the total at zero, the gun must gain an equal momentum in the opposite direction: $m_g v_g = -m_b v_b$. So the gun recoils backwards, and because it is much heavier, it moves much more slowly.

- Initial momentum zero: 1
- Equal and opposite momenta: 1
- Heavier gun, smaller recoil speed: 1

SECTION D

Q11. Answer the following: **[5]**

(i) The total momentum of two objects is unchanged by their collision if no external force acts. Proof: balls A and B collide for a time t. By the third law, $F_{AB} = -F_{BA}$, so $m_A(v_A - u_A)/t = -m_B(v_B - u_B)/t$. Rearranging: $m_A u_A + m_B u_B = m_A v_A + m_B v_B$, so the total momentum before equals the total momentum after. [3]

- Statement: 1
- Third-law forces in terms of momentum change: 1
- Final equality: 1

(ii) Momentum before = 0.01 × 300 + 0 = 3 kg m/s. After: (0.01 + 2) × v = 3, so v = 3/2.01 ≈ 1.49 m/s. [2]

- Conservation equation: 1
- v ≈ 1.49 m/s: 1

SECTION E

Q12. In a crash, a car stops almost instantly, but the passengers keep moving forward. Seat belts and airbags protect them. An airbag inflates in front of… **[4]**

(i) Newton's first law (inertia of motion). [1]

(ii) It increases the time taken to stop the passenger, so the force on the body is smaller (F = change in momentum/time). [1]

(iii) Change in momentum = 60 × 20 = 1200 kg m/s. (a) F = 1200/0.5 = 2400 N. (b) F = 1200/0.05 = 24,000 N, ten times larger. [2]

- Each force: 1